

NEW ZEALAND DATA SHEET

1 PRODUCT NAME

VERZENIO 50 mg film-coated tablets

VERZENIO 100 mg film-coated tablets

VERZENIO 150 mg film-coated tablets

2 QUALITATIVE AND QUANTITATIVE COMPOSITION

Abemaciclib 50 mg film-coated tablets

Each film-coated tablet contains 50 mg abemaciclib.

Excipients with known effect

Each film-coated tablet contains 14 mg of lactose (monohydrate).

Abemaciclib 100 mg film-coated tablets

Each film-coated tablet contains 100 mg abemaciclib.

Excipients with known effect

Each film-coated tablet contains 28 mg of lactose (monohydrate).

Abemaciclib 150 mg film-coated tablets

Each film-coated tablet contains 150 mg abemaciclib.

Excipients with known effect

Each film-coated tablet contains 42 mg of lactose (monohydrate).

For the full list of excipients, see Section [6.1 List of excipients](#).

3 PHARMACEUTICAL FORM

VERZENIO 50 mg film-coated tablets

Beige, modified oval tablet debossed with “Lilly” on one side and “50” on the other.

VERZENIO 100 mg film-coated tablets

White, modified oval tablet debossed with “Lilly” on one side and “100” on the other.

VERZENIO 150 mg film-coated tablets

Yellow, modified oval tablet debossed with “Lilly” on one side and “150” on the other.

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4 CLINICAL PARTICULARS

4.1 THERAPEUTIC INDICATIONS

Early breast cancer

VERZENIO in combination with endocrine therapy is indicated for the adjuvant treatment of adult patients 18 years of age or older with hormone receptor (HR) positive, human epidermal growth factor receptor 2 (HER2) negative, node-positive early breast cancer at high risk of recurrence.

In pre- or peri-menopausal women, endocrine therapy should be combined with a luteinising hormone-releasing hormone (LHRH) agonist.

Advanced or metastatic breast cancer

VERZENIO is indicated for the treatment of adult patients 18 years of age or older with hormone receptor (HR) positive, human epidermal growth factor receptor 2 (HER2) negative locally advanced or metastatic breast cancer in combination with an aromatase inhibitor or fulvestrant as initial endocrine-based therapy, or following prior endocrine therapy.

In pre- or peri-menopausal women, the endocrine therapy should be combined with a luteinising hormone-releasing hormone (LHRH) agonist.

4.2 DOSE AND METHOD OF ADMINISTRATION

VERZENIO therapy should be initiated and supervised by physicians experienced in the use of anti-cancer therapies.

The recommended dose of VERZENIO is 150 mg orally, twice daily in combination with endocrine therapy at its recommended dose.

VERZENIO may be taken with or without food. Avoid grapefruit or grapefruit juice.

Early breast cancer

Treatment with VERZENIO plus endocrine therapy should be combined with LHRH agonist for pre-menopausal women.

VERZENIO should be taken continuously for two years, or until disease recurrence or unacceptable toxicity occurs whichever comes first.

Advanced or metastatic breast cancer

Women treated with the combination of VERZENIO plus endocrine therapy should be in a postmenopausal state prior to therapy.

It is recommended that treatment be continued until disease progression or unacceptable toxicity.

Dose adjustments for managing adverse effects

Management of some adverse reactions may require dose interruption and/or dose reduction. If dose reduction is necessary, decrease the dose by 50 mg at a time. Discontinue VERZENIO for

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patients unable to tolerate 50 mg twice daily (see Section [4.4 Special warnings and precautions for use](#)).

Table 1. Recommended dose reductions for adverse reactions

Dose level	VERZENIO dose (in combination therapy)
Recommended starting dose	150 mg twice daily
First dose reduction	100 mg twice daily
Second dose reduction	50 mg twice daily

Table 2. Dose adjustment to manage haematological toxicities including neutropenia

Assess full blood counts prior to the start of VERZENIO therapy, every two weeks for the first two months, monthly for the next two months, and as clinically indicated.

CTCAE Grade	VERZENIO dose adjustments
Grade 1 or 2	No dose adjustment required.
Grade 3	Suspend dose until toxicity resolves to Grade 2 or less. Dose reduction is not required.
Grade 3, recurrent; or Grade 4	Suspend dose until toxicity resolves to Grade 2 or less. Resume at next lower dose.
Patient requires administration of blood cell growth factors	Suspend abemaciclib dose for at least 48 hours after the last dose of blood cell growth factors was administered and until toxicity resolves to Grade 2 or less. Resume at next lower dose unless the dose was already reduced for the toxicity that led to the use of the growth factor.

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Table 3. Dose adjustment to manage diarrhoea

At the first sign of loose stools, start treatment with anti-diarrhoeal agents, such as loperamide.

CTCAE Grade	VERZENIO dose adjustments
Grade 1	No dose adjustment required.
Grade 2	If toxicity does not resolve within 24 hours to Grade 1 or less, suspend dose until resolution. Dose reduction is not required.
Grade 2 that persists or recurs after resuming the same dose despite maximal supportive measures	Suspend dose until toxicity resolves to Grade 1 or less. Resume at next lower dose.
Grade 3 or 4 or requires hospitalisation	

Table 4. Dose adjustment to manage increased hepatotoxicity

Assess ALT/AST prior to the start of VERZENIO therapy, every two weeks for the first two months, monthly for the next two months, and as clinically indicated.

CTCAE Grade	VERZENIO dose adjustments
Grade 1 (>ULN-3.0 x ULN) Grade 2 (>3.0-5.0 x ULN)	No dose adjustment required.
Persistent or Recurrent Grade 2, or Grade 3 (>5.0-20.0 x ULN)	Suspend dose until toxicity resolves to baseline or Grade 1. Resume at next lower dose.
Elevation in AST and/or ALT >3 x ULN WITH total bilirubin >2 x ULN, in the absence of cholestasis	Discontinue abemaciclib.
Grade 4 (>20.0 x ULN)	Discontinue abemaciclib.

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Table 5. Dose adjustment to manage interstitial lung disease (ILD)/pneumonitis

CTCAE Grade	VERZENIO dose adjustments
Grade 1 or 2	No dose modification is required.
Grade 2 that persists or recurs despite maximal supportive measures and does not return to baseline or Grade 1 within 7 days	Suspend dose until toxicity resolves to baseline or ≤Grade 1. Resume at next lower dose.
Grade 3 or 4	Discontinue abemaciclib.

Table 6. Dose adjustment to manage venous thromboembolic events (VTEs)

CTCAE Grade	VERZENIO dose adjustments
Early breast cancer setting	
Any grade	Suspend dose and treat as clinically indicated. Abemaciclib may be resumed when the patient is clinically stable.
Metastatic breast cancer setting	
Grade 1 or 2	No dose modification is required.
Grade 3 or 4	Suspend dose and treat as clinically indicated. Abemaciclib may be resumed when the patient is clinically stable.

Table 7. Dose adjustment for other* toxicities

CTCAE Grade	VERZENIO dose adjustments
Grade 1 or 2	No dose adjustment required.
Persistent or recurrent Grade 2 toxicity that does not resolve with maximal supportive measures to baseline or Grade 1 within 7 days	Suspend dose until toxicity resolves to Grade 1 or less. Resume at next lower dose.
Grade 3 or 4	

* Toxicities other than diarrhoea, haematologic toxicity, hepatotoxicity, ILD/pneumonitis, and VTEs.

Dose adjustment related to drug-drug interactions

CYP3A inhibitors

Avoid concomitant use of strong CYP3A inhibitors (for example, voriconazole) and use caution with coadministration of moderate (for example, ciprofloxacin) or weak (for example, ranitidine)

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CYP3A inhibitors (see Section [4.5 Interaction with other medicines and other forms of interaction](#)).

If coadministration with a CYP3A inhibitor is unavoidable, adjust the abemaciclib dose as described in [Table 8](#).

Table 8. Dose modification in combination with CYP3A inhibitors^a

CYP3A inhibitor	Expected increase in exposure	VERZENIO dose recommendation
Specific inhibitors^b		
Ketoconazole	6.87 fold	50 mg once daily
Itraconazole	3.78 fold	50 mg twice daily
Clarithromycin	2.19 fold	100 mg twice daily
Diltiazem	2.41 fold	100 mg twice daily
Verapamil	1.63 fold	100 mg twice daily
For other inhibitors		
Strong inhibitor		50 mg twice daily

^a Based on a 150 mg twice daily starting dose.

^b Based on clinical results and physiologically-based pharmacokinetic simulations

In patients who have had a dose reduction to 50 mg twice daily due to adverse reactions, and in whom co-administration of a strong CYP3A4 inhibitor cannot be avoided, treatment with 50 mg twice daily may be continued with close monitoring of signs of toxicity. Alternatively, the dose may be reduced to 50 mg once daily or discontinued.

With concomitant use of moderate CYP3A inhibitors, monitor for adverse reactions and consider reducing the VERZENIO dose in 50 mg decrements as demonstrated in [Table 1](#), if necessary.

Avoid foods that inhibit CYP3A activity, such as grapefruit or grapefruit juice.

If a CYP3A inhibitor is discontinued, increase the VERZENIO dose (after 3-5 half-lives of the inhibitor) to the dose that was used before starting the inhibitor (see Section [4.5 Interaction with other medicines and other forms of interaction](#)).

CYP3A inducers

Avoid concomitant use of CYP3A inducers. Consider alternative agents without CYP3A induction (see Section [4.5 Interaction with other medicines and other forms of interaction](#)).

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Dosing in severe hepatic impairment

Decrease the dosing frequency to once daily for patients with severe hepatic impairment (see Section [5.2 Pharmacokinetic properties](#)). Dose adjustment is not required for patients with mild or moderate hepatic impairment.

4.3 CONTRAINDICATIONS

Hypersensitivity to the active substance or to any of the excipients listed in Section [6.1 List of excipients](#).

4.4 SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Neutropenia

Grade ≥ 3 neutropenia was reported in patients receiving abemaciclib in breast cancer studies. Monitor complete blood counts prior to starting abemaciclib therapy, every 2 weeks for the first 2 months, monthly for the next 2 months, and as clinically indicated. Fatal events of neutropenic sepsis occurred in $<1\%$ of patients with metastatic breast cancer. Patients should be instructed to report any episode of fever to their healthcare provider. Dose modification is recommended for patients who develop Grade 3 or 4 neutropenia (see Section [4.2 Dose and method of administration](#)).

Infections/infestations

Infections were reported in patients receiving abemaciclib plus endocrine therapy at a higher rate than in patients treated with endocrine therapy. Lung infection was reported in patients receiving abemaciclib without concurrent neutropenia. Fatal events occurred in $<1\%$ of patients with metastatic breast cancer. Patients should be monitored for signs and symptoms of infection and treated as medically appropriate.

Venous thromboembolism

VERZENIO has not been studied in patients with early breast cancer who had a history of venous thromboembolism. In early stage breast cancer, venous thromboembolisms of any grade were reported in 2.5% of patients treated with adjuvant abemaciclib plus endocrine therapy and 0.6% with endocrine therapy alone. Careful consideration should be given to the choice of endocrine therapy and known risks associated with VTEs, such as prior history of venous thromboembolic events.

In metastatic breast cancer venous thromboembolic events were reported in 5.3% of patients treated with abemaciclib plus fulvestrant or aromatase inhibitors, compared to 0.8% of patients treated with placebo plus fulvestrant or aromatase inhibitors.

Patients should be monitored for signs and symptoms of deep vein thrombosis and pulmonary embolism and treated as medically appropriate. Abemaciclib dose modification and suspension may be required (see Section [4.2 Dose and method of administration](#)).

Arterial thromboembolic events

Serious arterial thromboembolic events (ATEs), including ischemic stroke and myocardial infarction, have been reported more frequently in the abemaciclib treatment arms in metastatic breast cancer (mBC) studies when administered in combination with endocrine therapies.

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In mBC, ATEs of any grade have been reported in 4.4% of patients treated with abemaciclib plus endocrine therapy, compared to 2.4% of patients treated with placebo plus endocrine therapy. A causal association has not been established. The benefits and risks of continuing abemaciclib in patients who experience a severe ATE should be considered.

Increased ALT/AST

Grade ≥ 3 increased ALT/AST was reported in patients receiving abemaciclib in breast cancer studies. Monitor ALT/AST prior to the start of abemaciclib therapy, every 2 weeks for the first 2 months, monthly for the next 2 months, and as clinically indicated. Based on the level of ALT/AST elevations, dose modification may be required (see Section [4.2 Dose and method of administration](#)).

Interstitial lung disease (ILD)/pneumonitis

Severe, life threatening or fatal interstitial lung disease (ILD) and/or pneumonitis can occur in patients treated with VERZENIO and other CDK4/6 inhibitors. In VERZENIO treated patients in early breast cancer (monarchE), 3.2% of patients experienced ILD/pneumonitis of any grade: 0.4% were Grade 3 or 4 and there was one fatality ($<0.1\%$). In VERZENIO treated patients in metastatic breast cancer (MONARCH 1, MONARCH 2, MONARCH 3), 3.3% of VERZENIO-treated patients had ILD/pneumonitis of any grade, 0.6% had grade 3 or 4, and 0.4% had fatal outcomes. Additional cases of ILD/pneumonitis have been observed in the postmarketing setting, with fatalities reported.

Monitor patients for pulmonary symptoms indicative of ILD/pneumonitis. Symptoms may include hypoxia, cough, dyspnoea, or interstitial infiltrates on radiologic exams. Infectious, neoplastic and other causes for such symptoms should be excluded by means of appropriate investigations.

Dose interruptions or dose reduction is recommended for patients who develop persistent or recurrent Grade 2 ILD/pneumonitis. Permanently discontinue VERZENIO in all patients with Grade 3 or 4 ILD or pneumonitis (see Section [4.2 Dose and method of administration](#) for dose modification).

Diarrhoea

Diarrhoea is the most common adverse reaction. Across clinical studies, median time to onset of the first diarrhoea event was approximately 6 to 8 days, and median duration of diarrhoea was 7 to 12 days (Grade 2) and 5 to 8 days (Grade 3). Diarrhoea can be associated with dehydration. Patients should start treatment with anti-diarrhoeal agents such as loperamide at the first sign of loose stools, increase oral fluids and notify their healthcare provider. Dose modification is recommended for patients who develop $\geq$ Grade 2 diarrhoea (see Section [4.2 Dose and method of administration](#)).

Use in hepatic impairment

Abemaciclib is metabolised in the liver. Mild and moderate hepatic impairment (Child-Pugh classification), did not have a clinically relevant effect on abemaciclib exposure. In subjects with severe hepatic impairment (Child-Pugh classification), total abemaciclib unbound exposure increased 2.69-fold, and the abemaciclib half-life increased from 24 to 55 hours. Reduce the abemaciclib dosing frequency to once daily in patients with severe hepatic impairment.

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Use in renal impairment

Abemaciclib and its metabolites are not significantly cleared renally. Dose adjustment is not necessary in patients with mild or moderate renal impairment. There are no data in patients with severe renal impairment, end stage renal disease, or in patients on dialysis.

Paediatric use

The safety and efficacy of abemaciclib in children aged less than 18 years has not been established. No data are available.

Lactose

Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine.

4.5 INTERACTION WITH OTHER MEDICINES AND OTHER FORMS OF INTERACTION

Effects of other medicinal products on the pharmacokinetics of abemaciclib

Abemaciclib is primarily metabolised by CYP3A4.

CYP3A4 inhibitors

Co-administration of abemaciclib with CYP3A4 inhibitors can increase plasma concentrations of abemaciclib. In patients with advanced and/or metastatic cancer, co-administration of the CYP3A4 inhibitor clarithromycin resulted in a 3.4-fold increase in the plasma exposure of abemaciclib and a 2.5-fold increase in the combined unbound potency adjusted plasma exposure of abemaciclib and its active metabolites.

Use of strong CYP3A4 inhibitors together with abemaciclib should be avoided. If strong CYP3A4 inhibitors need to be co-administered, the dose of abemaciclib should be reduced (see Section [4.2 Dose and method of administration](#); [Table 8](#)), followed by careful monitoring of toxicity. Examples of strong CYP3A4 inhibitors include, but not limited to: clarithromycin, itraconazole, ketoconazole, lopinavir/ritonavir, posaconazole or voriconazole. Avoid grapefruit or grapefruit juice.

The dose of abemaciclib should be reduced if co-administered with moderate CYP3A4 inhibitors verapamil and diltiazem (see Section [4.2 Dose and method of administration](#); [Table 8](#)).

No dose adjustment is necessary for patients treated with other moderate or weak CYP3A4 inhibitors. There should, however, be close monitoring for signs of toxicity.

CYP3A4 inducers

Co-administration of abemaciclib with the strong CYP3A4 inducer rifampicin decreased the plasma concentration of abemaciclib by 95% and unbound potency adjusted plasma concentration of abemaciclib plus its active metabolites by 77% based on AUC_{0-∞}. Concomitant use of strong CYP3A4 inducers (including, but not limited to: carbamazepine, phenytoin, rifampicin and St. John's wort) should be avoided due to the risk of decreased efficacy of abemaciclib.

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Effects of abemaciclib on the pharmacokinetics of other medicinal products

Medicinal products that are substrates of transporters

Abemaciclib and its major active metabolites inhibit the renal transporters organic cation transporter 2 (OCT2), multidrug and extrusion toxin protein (MATE1), and MATE2-K. *In vivo* interactions of abemaciclib with clinically relevant substrates of these transporters, such as dofetilide or creatinine, may occur (see Section 4.8 Undesirable effects). In a clinical drug interaction study with metformin (substrate of OCT2, MATE1 and 2) co-administered with 400 mg abemaciclib, a small but not clinically relevant increase (37%) in metformin plasma exposure was observed. This was found to be due to reduced renal secretion with unaffected glomerular filtration.

In healthy subjects, co-administration of abemaciclib and the P-glycoprotein (P-gp) substrate loperamide resulted in an increase in loperamide plasma exposure of 9% based on $AUC_{0-\infty}$ and 35% based on C_{max} . This was not considered to be clinically relevant. However, based on the *in vitro* inhibition of P-gp and breast cancer resistance protein (BCRP) observed with abemaciclib, *in vivo* interactions of abemaciclib with narrow therapeutic index substrates of these transporters, such as digoxin or dabigatran etexilate, may occur.

In a clinical study in patients with breast cancer, there was no clinically-relevant pharmacokinetic drug interaction between abemaciclib and anastrozole, fulvestrant, exemestane, letrozole or tamoxifen. It is currently unknown whether abemaciclib may reduce the effectiveness of systemically acting hormonal contraceptives, and therefore patients using systemically acting hormonal contraceptives are advised to add a barrier method.

4.6 FERTILITY, PREGNANCY AND LACTATION

Fertility

Effects on fertility and early embryonic development were assessed in rats. While no effects on male fertility were seen in rats with abemaciclib up to 10 mg/kg/day (approximately 4 times higher than clinical exposure based on AUC), effects to the male reproductive organs in mice, rats and dogs (see below) indicate that abemaciclib may impair fertility in males. No effects on female fertility and early embryonic development were observed in rats at up to 20 mg/kg/day (approximately 4.5 times the clinical exposure based on AUC).

In toxicity studies in mice (150 mg/kg/day), rats (≥ 0.3 mg/kg/day) and dogs (≥ 0.3 mg/kg/day), abemaciclib-related findings in the testis, epididymis, prostate, and seminal vesicle included decreased organ weights, intratubular cellular debris, hypospermia, tubular dilatation, atrophy, and/or degeneration/necrosis. Exposures at these doses in rats and dogs are less than the clinical exposure based on AUC and 11 times higher in mice than clinical exposure. In female mice, decreased number of corpora lutea were observed at ≥ 30 mg/kg/day, which is approximately 7.9 times higher than clinical exposure based.

Patients who could become pregnant should use highly effective contraception during treatment and for 3 weeks after the last dose of abemaciclib.

Pregnancy

There are no data on the use of VERZENIO in human pregnancy. Based on findings in animals, and its mechanism of action, abemaciclib can cause fetal harm when administered to a pregnant patient. When pregnant rats were treated during the period of organogenesis

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(dose of ≥ 4 mg/kg/day; approximately equal to the human clinical exposure based on AUC), reduced fetal weights were observed in the absence of maternal toxicity, accompanied by an increased incidence of cardiovascular and skeletal malformations and variations (absent innominate artery and aortic arch, malpositioned subclavian artery, unossified sternbra, bipartite ossification of thoracic centrum, and rudimentary or nodulated ribs).

VERZENIO is not recommended during pregnancy. Highly effective contraception is recommended.

Lactation

There are no data on the presence of abemaciclib in human milk, effects of abemaciclib on the breastfed child, or effects of abemaciclib on milk production. Due to the potential for serious adverse reactions in a child, patients should not breastfeed during treatment with abemaciclib, or for three weeks after the last dose of abemaciclib.

4.7 EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

The effects of VERZENIO on a person's ability to drive and use machines were not assessed as part of its registration.

4.8 UNDESIRABLE EFFECTS

Reporting suspected adverse reactions after authorisation of the medicine is important. It allows continued monitoring of the benefit/risk balance of the medicine. Healthcare professionals are asked to report any suspected adverse reactions <https://pophealth.my.site.com/carmreportnz/s/>

Summary of the safety profile

The most commonly occurring adverse reactions are diarrhoea, infections, neutropenia, leukopenia, anaemia, fatigue, nausea, vomiting, decreased appetite and alopecia.

Safety in early breast cancer studies

monarchE – abemaciclib plus either tamoxifen or an aromatase inhibitor in patients with early breast cancer

The safety of VERZENIO in early breast cancer was evaluated in monarchE, a study in which 2791 adult patients received VERZENIO plus endocrine therapy (either tamoxifen or an aromatase inhibitor) and 2800 received endocrine therapy alone (see section 5.1 Pharmacodynamic properties – Clinical efficacy).

The most frequently ($\geq 5\%$) reported Grade ≥ 3 adverse reactions were neutropenia, leukopenia, diarrhoea and lymphopenia.

Fatal adverse reactions occurred in 0.8% of patients who received VERZENIO plus endocrine therapy including: cardiac failure (0.1%), cardiac arrest, myocardial infarction, ventricular fibrillation, cerebral haemorrhage, cerebrovascular accident, pneumonitis, hypoxia, diarrhoea and mesenteric artery thrombosis (0.03% each).

VERZENIO treatment discontinuation due to an adverse reaction was reported in 19% of patients receiving VERZENIO, plus endocrine therapy. The most common adverse reactions leading to

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VERZENIO discontinuations were diarrhoea (5%), fatigue (2%), and neutropenia (0.9%). Of the patients receiving endocrine therapy, 1% permanently discontinued due to an adverse reaction.

Dose interruption of VERZENIO due to an adverse reaction occurred in 62% of patients receiving VERZENIO plus endocrine therapy. Adverse reactions leading to VERZENIO dose interruptions in $\geq 3\%$ of patients were diarrhoea (20%), neutropenia (16%), leukopenia (7%), and fatigue (5%).

Dose reductions of VERZENIO due to adverse reaction occurred in 44% of patients receiving VERZENIO plus endocrine therapy. Adverse reactions leading to VERZENIO dose reductions were diarrhoea (17%), neutropenia (8%), fatigue (5%) and leukopenia (4%).

The most common adverse events and laboratory abnormalities with abemaciclib treatment in monarchE are summarised in [Table 9](#) and [Table 10](#), respectively.

Table 9. Adverse events ($\geq 10\%$) of patients receiving abemaciclib plus tamoxifen or an aromatase inhibitor [with a difference between arms of $\geq 2\%$] in monarchE

	Abemaciclib plus either tamoxifen or an aromatase inhibitor (investigational arm) N = 2791			Tamoxifen or an aromatase inhibitor (comparator arm) N = 2800		
	All Grades ^a %	Grade 3 %	Grade 4 %	All Grades ^b %	Grade 3 %	Grade 4 %
Gastrointestinal disorders						
Diarrhoea	84	8	0	9	0.2	0
Nausea	30	0.5	0	9	<0.1	0
Vomiting	18	0.5	0	5	0.1	0
Stomatitis ^c	14	0.1	0	5	0	0
Infections and infestations						
Infections ^d	51	4.9	0.6	39	2.7	0.1
General disorders and administration site conditions						
Fatigue ^e	41	2.9	0	18	0.1	0
Nervous system disorders						
Headache	20	0.3	0	15	0.2	0
Dizziness	11	0.1	0	7	<0.1	0
Metabolism and nutrition disorders						
Decreased appetite	12	0.6	0	2	<0.1	0
Skin and subcutaneous tissue disorders						
Rash ^f	11	0.4	0	5	0	0
Alopecia	11	0	0	3	0	0

^a Includes the following fatal adverse reactions: diarrhoea (n=1), and infections (n=4)

^b Includes the following fatal adverse reactions: infections (n=5)

^c Includes mouth ulceration, mucosal inflammation, oropharyngeal pain, stomatitis.

^d Includes all reported preferred terms that are part of the Infections and Infestations system organ class. Most common infections (>5%) include upper respiratory tract infection, urinary tract infection, and nasopharyngitis.

^e Includes asthenia, fatigue.

^f Includes exfoliative rash, mucocutaneous rash, rash, rash erythematous, rash follicular, rash generalized, rash macular, rash maculopapular, rash maculovesicular, rash morbilliform, rash papular, rash papulosquamous, rash pruritic, rash vesicular, vulvovaginal rash.

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Clinically relevant adverse events in <10% of patients who received VERZENIO in combination with tamoxifen or an aromatase inhibitor in monarchE include:

- Pruritus - 9%
- Dyspepsia - 8%
- Nail disorder - 6% (includes nail bed disorder, nail bed inflammation, nail discoloration, nail disorder, nail dystrophy, nail pigmentation, nail ridging, nail toxicity, onychalgia, onychoclasia, onycholysis, onychomadesis)
- Lacrimation increased - 6%
- Dysgeusia - 5%
- Interstitial lung disease (ILD)/pneumonitis - 3% (includes pneumonitis, radiation pneumonitis, interstitial lung disease, pulmonary fibrosis, organising pneumonia, radiation fibrosis – lung, lung opacity, sarcoidosis)
- Venous thromboembolic events (VTEs) - 3% (includes catheter site thrombosis, cerebral venous thrombosis, deep vein thrombosis, device related thrombosis, embolism, hepatic vein thrombosis, jugular vein occlusion, jugular vein thrombosis, ovarian vein thrombosis, portal vein thrombosis, pulmonary embolism, subclavian vein thrombosis, venous thrombosis limb)

Table 10. Laboratory abnormalities (≥10%) in patients receiving abemaciclib plus tamoxifen or an aromatase inhibitor [with a difference between arms of ≥2%] in monarchE

	Abemaciclib plus either tamoxifen or an aromatase inhibitor (investigational arm) N = 2791			Tamoxifen or an aromatase inhibitor (comparator arm) N = 2800		
	All Grades %	Grade 3 %	Grade 4 %	All Grades %	Grade 3 %	Grade 4 %
Creatinine increased	99	0.5	0	91	<0.1	0
White blood cell decreased	89	19	<0.1	28	1.1	0
Neutrophil count decreased	84	18	0.7	23	1.6	0.3
Anaemia	68	1.0	0	17	0.1	0
Lymphocyte count decreased	59	13	0.2	24	2.4	0.1
Platelet count decreased	37	0.7	0.2	10	0.1	0.1
ALT increased	37	2.5	<0.1	24	1.2	0
AST increased	31	1.5	<0.1	18	0.9	0
Hypokalaemia	11	1.2	0.1	3.8	0.1	0.1

ALT = alanine aminotransferase; AST = aspartate aminotransferase

Safety in advanced or metastatic breast cancer studies

The safety of VERZENIO in advanced or metastatic breast cancer was evaluated in MONARCH 3 and MONARCH 2 (see Section [5.1 Pharmacodynamic properties](#) – Clinical efficacy).

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MONARCH 3 – abemaciclib plus aromatase inhibitor in postmenopausal women with HR-positive, HER2-negative locoregionally recurrent or metastatic breast cancer with no prior systemic therapy in this disease setting

In MONARCH 3, 488 women were randomly assigned to receive 150 mg of VERZENIO or placebo orally twice daily, plus physician's choice of anastrozole or letrozole once daily (see Section 5.1 Pharmacodynamic properties – Clinical trials). Median duration of treatment was 15 months for the VERZENIO arm and approximately 14 months for the placebo arm.

The most frequently reported ($\geq 5\%$) Grade 3 or 4 adverse reactions were neutropenia, diarrhoea, leukopenia, increased ALT, and anaemia.

Deaths during treatment or during the 30-day follow up, regardless of causality, were reported in 11 cases (3%) of VERZENIO plus an aromatase inhibitor treated patients versus 3 cases (2%) of placebo plus an aromatase inhibitor treated patients. Causes of death for patients receiving VERZENIO plus an aromatase inhibitor included: 3 (0.9%) patient deaths due to underlying disease, 3 (0.9%) due to lung infection, 3 (0.9%) due to VTE, 1 (0.3%) due to pneumonitis, and 1 (0.3%) due to cerebral infarction.

Permanent treatment discontinuation due to an adverse reaction was reported in 13% of patients receiving VERZENIO plus an aromatase inhibitor and in 3% of patients receiving placebo plus an aromatase inhibitor. Adverse reactions leading to permanent discontinuation for patients receiving VERZENIO plus an aromatase inhibitor were diarrhoea (2%), ALT increased (2%), infection (1%), venous thromboembolic events (VTE) (1%), neutropenia (0.9%), renal impairment (0.9%), AST increased (0.6%), dyspnoea (0.6%), pulmonary fibrosis (0.6%) and anaemia, rash, weight decreased and thrombocytopenia (each 0.3%).

Dose interruption of VERZENIO due to an adverse reaction occurred in 56% of patients receiving VERZENIO plus anastrozole or letrozole. Adverse reactions leading to VERZENIO dose interruptions in $\geq 5\%$ of patients were neutropenia (16%) and diarrhoea (15%).

Dose reductions due to an adverse reaction occurred in 43% of patients receiving VERZENIO plus anastrozole or letrozole. Adverse reactions leading to dose reductions in $\geq 5\%$ of patients were diarrhoea and neutropenia. VERZENIO dose reductions due to diarrhoea of any grade occurred in 13% of patients receiving VERZENIO plus an aromatase inhibitor compared to 2% of patients receiving placebo plus an aromatase inhibitor. VERZENIO dose reductions due to neutropenia of any grade occurred in 11% of patients receiving VERZENIO plus an aromatase inhibitor compared to 0.6% of patients receiving placebo plus an aromatase inhibitor.

The most common adverse events are shown in Table 11. The most common laboratory abnormalities in patients receiving VERZENIO in MONARCH 3 were similar to those in monarchE, in the early breast cancer setting (see Table 10).

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Table 11. Adverse events (≥10%) in patients receiving abemaciclib plus anastrozole or letrozole [with a difference between arms of ≥2%] in MONARCH 3

	Abemaciclib plus either anastrozole or letrozole (investigational arm) N=327			Placebo plus either anastrozole or letrozole (comparator arm) N=161		
	All Grades (%)	Grade 3 (%)	Grade 4 (%)	All Grades (%)	Grade 3 (%)	Grade 4 (%)
Gastrointestinal disorders						
Diarrhoea	81	9	0	30	1.2	0
Nausea	39	0.9	0	20	1.2	0
Abdominal pain	29	1.2	0	12	1.2	0
Vomiting	28	1.2	0	12	1.9	0
Constipation	16	0.6	0	12	0	0
Infections and infestations						
Infections ^a	39	4.0	0.9	29	2.5	0.6
General disorders and administration site conditions						
Fatigue	40	1.8	0	32	0	0
Influenza like illness	10	0	0	8	0	0
Skin and subcutaneous tissue disorders						
Alopecia	27	0	0	11	0	0
Rash	14	0.9	0	5	0	0
Pruritus	13	0	0	9	0	0
Metabolism and nutrition disorders						
Decreased appetite	24	1.2	0	9	0.6	0
Investigations						
Weight decreased	10	0.6	0	3	0.6	0
Respiratory, thoracic and mediastinal disorders						
Cough	13	0	0	9	0	0
Dyspnoea	12	0.6	0.3	6	0.6	0
Nervous system disorders						
Dizziness	11	0.3	0	9	0	0

^a Includes all reported preferred terms that are part of the *Infections and infestations* system organ class. Most common infections (>1%) include upper respiratory tract infection, lung infection, and pharyngitis.

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MONARCH 2 - abemaciclib plus fulvestrant in women with HR-positive, HER2-negative advanced or metastatic breast cancer with disease progression on or after prior adjuvant or metastatic endocrine therapy

In MONARCH 2, women were randomised to receive fulvestrant plus either abemaciclib or placebo (see Section 5.1 Pharmacodynamic properties – Clinical efficacy). The overall safety profile of abemaciclib was broadly similar in MONARCH 2 and MONARCH 3.

In MONARCH 2, deaths during treatment or during the 30-day follow up period, regardless of causality, were reported for 18 patients (4%) receiving VERZENIO plus fulvestrant and 10 patients (5%) receiving placebo plus fulvestrant. Causes of death for patients receiving VERZENIO plus fulvestrant included: 7 (2%) patient deaths due to underlying disease, 4 (0.9%) due to sepsis, 2 (0.5%) due to pneumonitis, 2 (0.5%) due to hepatotoxicity, and one (0.2%) due to cerebral infarction.

The most common adverse events in MONARCH 2 are summarised in Table 12. The most common laboratory abnormalities in patients receiving VERZENIO in MONARCH 2 were similar to those in monarchE, in the early breast cancer setting (see Table 10).

Table 12. Adverse events (≥10%) in patients receiving abemaciclib plus fulvestrant [with a difference between arms of ≥2%] in MONARCH 2

	Abemaciclib plus fulvestrant (investigational arm) N=441			Placebo plus fulvestrant (comparator arm) N=223		
	All Grades (%)	Grade 3 (%)	Grade 4 (%)	All Grades (%)	Grade 3 (%)	Grade 4 (%)
Gastrointestinal disorders						
Diarrhoea	86	13	0	25	0.4	0
Nausea	45	2.7	0	23	0.9	0
Abdominal pain ^a	35	2.5	0	16	0.9	0
Vomiting	26	0.9	0	10	1.8	0
Stomatitis	15	0.5	0	10	0	0
Infections and infestations						
Infections ^b	43	5	0.7	25	3.1	0.4
General disorders and administration site conditions						
Fatigue ^c	46	2.7	0	32	0.4	0
Oedema peripheral	12	0	0	7	0	0
Pyrexia	11	0.5	0.2	6	0.4	0
Metabolism and nutrition disorders						
Decreased appetite	27	1.1	0	12	0.4	0

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	Abemaciclib plus fulvestrant (investigational arm) N=441			Placebo plus fulvestrant (comparator arm) N=223		
	All Grades (%)	Grade 3 (%)	Grade 4 (%)	All Grades (%)	Grade 3 (%)	Grade 4 (%)
Respiratory, thoracic and mediastinal disorders						
Cough	13	0	0	11	0	0
Skin and subcutaneous tissue disorders						
Alopecia	16	0	0	2	0	0
Pruritus	13	0	0	6	0	0
Rash	11	1.1	0	5	0	0
Nervous system disorders						
Headache	20	0.7	0	15	0.4	0
Dysgeusia	18	0	0	3	0	0
Dizziness	12	0.7	0	6	0	0
Investigations						
Weight decreased	10	0.2	0	2	0.4	0

^a Includes abdominal pain, abdominal pain upper, abdominal pain lower, abdominal discomfort, abdominal tenderness.

^b Includes upper respiratory tract infection, urinary tract infection, lung infection, pharyngitis, conjunctivitis, sinusitis, vaginal infection, sepsis.

^c Includes asthenia, fatigue.

Additional adverse events of note in MONARCH 2 included venous thromboembolic events (deep vein thrombosis, pulmonary embolism, cerebral venous sinus thrombosis, subclavian vein thrombosis, axillary vein thrombosis, and DVT inferior vena cava), which were reported in 5% of patients treated with VERZENIO plus fulvestrant as compared to 0.9% of patients treated with fulvestrant plus placebo.

Description of selected adverse reactions

Neutropenia

In monarchE study, neutropenia was reported in 46 % of patients. Grade ≥3 decrease in neutrophil counts (based on laboratory findings) was reported in 19 % of patients receiving abemaciclib in combination with endocrine therapy. The median time to onset of Grade ≥3 neutropenia was 30 days, and median time to resolution was 16 days. Febrile neutropenia was reported in 0.3% patients. Neutropenia was also very common in MONARCH 2 and MONARCH 3 (45%) and a Grade 3 or 4 decrease in neutrophil counts (based on laboratory findings) was reported in 28% of patients receiving abemaciclib in combination with aromatase inhibitors or fulvestrant. The median time to onset of Grade 3 or 4 neutropenia were 29 and 33 days, and median time to resolution were 11 and 15 days. Febrile neutropenia was reported in 0.9% of patients. Dose modification is recommended for patients who develop Grade 3 or 4 neutropenia (see Section 4.2 Dose and method of administration).

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Diarrhoea

Diarrhoea was the most commonly reported adverse reaction (see [Table 9](#), [Table 11](#) and [Table 12](#)). Incidence was greatest during the first month of abemaciclib treatment. The median time to onset of a first diarrhoea event of any grade was approximately 6 to 8 days across studies, and the median duration of Grade 2 diarrhoea was 6 to 11 days across studies and of Grade 3 diarrhoea was 5 to 8 days across studies. Most diarrhoea events recovered or resolved with supportive treatment such as loperamide and/or dose adjustment (see [Section 4.2 Dose and method of administration](#)). In MONARCH 3, 19% of patients with diarrhoea required a dose omission and 13% required a dose reduction. The median time to the first dose reduction due to diarrhoea in MONARCH 3 was 38 days.

Increased aminotransferases (ALT/AST)

Dose modification is recommended for patients who develop Grade 3 or 4 ALT or AST increase (see [Section 4.2 Dose and method of administration](#)).

In the monarchE study, ALT and AST elevations were very common in patients receiving abemaciclib in combination with endocrine therapy (see [Table 10](#)). Grade 3 - 4 ALT and AST elevations (based on laboratory findings) were reported in 2.6 % and 1.6 % of patients, respectively. The median time to onset of Grade ≥ 3 ALT elevation was 118 days, and median time to resolution was 15 days. The median time to onset of Grade ≥ 3 AST elevation was 91 days, and median time to resolution was 11 days.

ALT and AST elevations were also very common in the MONARCH 2 and MONARCH 3 studies amongst patients receiving abemaciclib in combination with aromatase inhibitors or fulvestrant. Grade 3 or 4 ALT or AST elevations (based on laboratory findings) were reported in 6.1% and 4.2% of patients, respectively. The median time to onset of Grade 3- 4 ALT elevation was 57 days in MONARCH 2 and 61 days in MONARCH 3, and median time to resolution was 14 days in both studies. The median time to onset of Grade 3 - 4 AST elevation was 71 days in MONARCH 3 and 185 days in MONARCH 2, and median time to resolution was 13 days (MONARCH 2) and 15 days (MONARCH 3).

Creatinine

Abemaciclib causes asymptomatic serum creatinine elevation by inhibiting renal tubular secretion transporters, but does not affect glomerular function as measured by iohexol clearance (see [Section 4.5 Interaction with other medicines and other forms of interaction](#)). In clinical studies, increases in serum creatinine occurred within the first month of abemaciclib dosing, remained elevated but stable through the treatment period, and were reversible upon treatment discontinuation. Creatinine elevations were not accompanied by changes in markers of renal function, such as blood urea nitrogen (BUN), cystatin C, or calculated glomerular filtration rate based on cystatin C.

Laboratory findings of serum creatinine elevation occurred in 99% of patients randomised to receive abemaciclib in the monarchE study, and 98% of those in the MONARCH 2 and MONARCH 3 studies. The incidence of Grade 3 or higher elevations was 0.5% and 2%, respectively. The incidence of laboratory findings of serum creatinine elevation amongst patients in the comparator arm was 91% in monarchE and 78% in MONARCH 2 and MONARCH 3.

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Other special populations

Asian patients

In MONARCH 2, the incidence of Grade ≥ 3 treatment-emergent adverse effects (TEAEs) was higher in Asian patients than in non-Asian (Caucasian and “other”) patients for neutropenia (44.6% and 17.6%, respectively), leukopenia (13.5% and 6.9%, respectively), anaemia (11.5% and 5.3%, respectively), thrombocytopenia (8.1% and 0.8%, respectively), increased ALT (6.8% and 2.7%, respectively), and increased AST (4.7% and 1.1%, respectively). No major difference in the incidence of diarrhoea was observed.

In MONARCH 3, the incidence of Grade ≥ 3 TEAEs was higher in Asian patients than in non-Asian (Caucasian and “other”) patients for neutropenia (27.2% and 17.3%, respectively), increased ALT (12.6% and 3.1%, respectively), anaemia (9.7% and 4.1%, respectively), and increased AST (7.8% and 1.5%, respectively). No major difference in the incidence of diarrhoea was observed.

Patients aged ≥ 65 years

In MONARCH 3, the incidence Grade ≥ 3 TEAEs was higher in patients aged ≥ 65 years than in those aged <65 years for neutropenia (23.6% and 19%, respectively), diarrhoea (12.2% and 7.3%, respectively), leukopenia (9.5% and 6.1%, respectively), anaemia (8.1% and 3.9%, respectively), and increased ALT (7.4% and 5%, respectively).

Post-marketing experience

The following adverse drug reactions are based on post-marketing reports.

Respiratory, thoracic, and mediastinal disorders: Interstitial lung disease/pneumonitis: Common ($\geq 1.0\%$ - $<10\%$)

Skin and subcutaneous tissue disorders: Erythema multiforme: rare ($\geq 0.01\%$ - $<0.1\%$)

4.9 OVERDOSE

There is no known antidote for abemaciclib overdose. In case of overdose, use supportive therapy.

For risk assessment and advice on the management of overdose please contact the National Poisons Centre on 0800 POISON (0800 764766).

5 PHARMACOLOGICAL PROPERTIES

5.1 PHARMACODYNAMIC PROPERTIES

Pharmacotherapeutic group: Antineoplastic agents, protein kinases inhibitors, ATC code: L01EF03

In cancer patients, abemaciclib inhibits CDK4 and CDK6 as indicated by inhibition of phosphorylation of Rb and topoisomerase II alpha, which results in cell cycle inhibition upstream of the G1 restriction point at doses of 50 mg to 200 mg twice daily. MONARCH 2 and MONARCH 3 exposure-response analyses support the 150-mg twice daily starting dose in combination with endocrine therapy and support dose reductions as needed for tolerability to a dose as low as 50 mg twice daily. The effect of abemaciclib on the QTcF interval was evaluated in 144 patients with advanced cancer. No large change (that is, >20 ms) in the QTcF interval was detected at the

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mean observed maximal steady state abemaciclib concentration following a therapeutic dosing schedule. In an exposure-response analysis in healthy subjects at the highest clinically relevant exposures, abemaciclib did not prolong the QTcF interval to any clinically relevant extent.

Mechanism of action

Abemaciclib is an inhibitor of cyclin D-dependent kinases 4 and 6 (CDK4 and CDK6) and was most active against cyclin D1/CDK4 in enzymatic assays. In breast cancer, cyclin D1/CDK4 has been shown to promote phosphorylation of the retinoblastoma protein (Rb), cell proliferation, and tumour growth. Abemaciclib prevents Rb phosphorylation, blocking progression from G1 into S phase of the cell cycle, leading to suppression of tumour growth in preclinical models following short duration target inhibition. In oestrogen receptor-positive breast cancer cell lines, sustained target inhibition by abemaciclib prevents rebound of Rb phosphorylation and cell cycle reentry, resulting in senescence and apoptosis. In breast cancer xenograft models, abemaciclib dosed daily without interruption at clinically relevant doses – as a single agent or in combination with antioestrogens – resulted in reduction of tumour size.

Clinical efficacy

Cardiac electrophysiology

The effect of abemaciclib on the QTcF interval was evaluated in 144 patients with advanced cancer. No large change (that is, >20 ms) in the QTcF interval was detected at the mean observed maximal steady state abemaciclib concentration following a therapeutic dosing schedule.

In an exposure-response analysis in healthy subjects at exposures comparable to a 200 mg twice-daily dose, abemaciclib did not prolong the QTcF interval to any clinically relevant extent.

Early breast cancer

Randomised phase 3 study monarchE: VERZENIO in combination with endocrine therapy

The efficacy and safety of VERZENIO in combination with adjuvant endocrine therapy was evaluated in monarchE a randomised, open-label, multicentre study in adult women and men with HR-positive, HER2-negative, node-positive, resected, early breast cancer (EBC) with clinical and pathological features consistent with a high risk of disease recurrence. To be enrolled, all patients had to have HR positive, HER2-negative EBC with tumour involvement in at least 1 axillary lymph node (pALN). Two cohorts of patients were enrolled. To be enrolled in cohort 1, patients needed to have either ≥ 4 pALN, or pALN 1-3 and either tumour grade 3 or tumour size ≥ 50 mm. To be enrolled in cohort 2, patients were required to have pALN 1-3 and Ki-67 index of $\geq 20\%$ as measured in untreated breast tumour tissue, using a clinical trial assay at a central laboratory. The intent to treat (ITT) population included patients from both cohort 1 (n=5120) and cohort 2 (n=517). Key exclusion criteria included inflammatory breast cancer, pre-existing conditions resulting in clinically significant diarrhoea (such as Crohn's Disease, ulcerative colitis, or major resection of the small bowel and stomach), history of cardiovascular issues in the last 12 months, including history of venous thromboembolism. Randomisation to treatment was stratified by prior treatment (neoadjuvant chemotherapy versus adjuvant chemotherapy versus no chemotherapy); menopausal status (premenopausal versus postmenopausal); and region (North America/Europe versus Asia versus other). Men were stratified as postmenopausal.

A total of 5637 patients were randomised in a ratio 1:1 to receive 2 years of VERZENIO 150 mg twice daily plus physician's choice of standard endocrine therapy or standard endocrine therapy

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alone; with or without LHRH agonist. After the end of the study treatment period, standard adjuvant endocrine therapy is to be continued for a duration of at least 5 years if deemed medically appropriate. Initial endocrine therapy received by patients included letrozole (38.2%), tamoxifen (31.4%), anastrozole (22.0%), or exemestane (8.1%). The use of LHRH analogues during the study was 21.7% and 22.4% in VERZENIO plus endocrine therapy and endocrine therapy arms, respectively. In premenopausal women, (N=2451), aromatase inhibitor and tamoxifen was used in 41.2% and 58.2% patients respectively, and LHRH agonists was used in 74.6% patients on aromatase inhibitor and in 30.1% patients on tamoxifen.

Patient median age was 51 years (range, 22-89 years), 99% were women, 71% were White, and 24% were Asian. Forty-four percent of patients were premenopausal. Most patients received prior chemotherapy (37% neoadjuvant, 62% adjuvant) and prior radiotherapy (95%). Sixty percent of the patients had 4 or more positive lymph nodes with 20% having ≥ 10 positive lymph nodes, 38% had Grade 3 tumour, and 22% had pathological tumour size ≥ 50 mm. Most patients were progesterone receptor positive (87%) and 44% had high Ki-67 index as determined by the central clinical trial assay.

The primary end point was invasive disease-free survival (IDFS). IDFS was defined as the time from randomisation to the first occurrence of ipsilateral invasive breast tumour recurrence, regional invasive breast cancer recurrence, distant recurrence, contralateral invasive breast cancer, second primary non-breast invasive cancer, or death attributable to any cause. Secondary end point distant relapse free survival (DRFS) is defined as the time from randomisation to distant recurrence or death from any cause, whichever occurs first.

At the pre-planned interim analysis (IA2) with a median follow-up time of 15.4 months, the primary objective of the study was met. A statistically significant improvement in IDFS was observed in patients who received VERZENIO plus endocrine therapy versus endocrine therapy alone (HR = 0.747, 95 % CI [0.598, 0.932], $p = 0.0096$). In addition, a clinically meaningful benefit in DRFS (HR = 0.717, 95 % CI [0.559, 0.920], nominal $p=0.00853$) was observed with VERZENIO plus endocrine therapy, reflecting a 28.3 % reduction in the risk of distant recurrence or death. Consistent results were observed in patient subgroups including geographic region, prior chemotherapy, and menopausal status.

At the time of the final overall survival (OS) analysis (15 July 2025 data cut-off), all patients had completed or discontinued the 2-year treatment period, and 87% had at least 5 years of follow-up. The median follow-up was 76 months (6.3 years) in both treatment arms. A statistically significant improvement in OS was observed in the ITT population, with VERZENIO plus endocrine therapy reducing the risk of death by 15.8% compared with endocrine therapy alone (HR = 0.842, 95% CI [0.722, 0.981], $p = 0.027$).

In the ITT population, VERZENIO plus endocrine therapy reduced the hazard of developing an IDFS event by 26.6% (HR = 0.734, 95% CI [0.657, 0.820], nominal $p<0.0001$) compared with endocrine therapy alone, corresponding to an absolute improvement of 6.6% in the 7-year IDFS rate. A clinically meaningful benefit in DRFS was observed (HR = 0.746, 95% CI [0.662, 0.840], nominal $p=0.0001$), reflecting a 25.4% reduction in the risk of a DRFS event.

The efficacy results for the final OS analysis are summarised in [Table 13](#).

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Table 13. monarchE: Efficacy Results at Final OS Analysis (Intent-to-Treat Population)

	VERZENIO Plus Endocrine Therapy N=2808	Endocrine Therapy Alone N=2829
Invasive Disease-Free Survival (IDFS)		
Number of patients with an event (n, %)	547 (19.5)	722 (25.5)
Hazard ratio (95% CI)	0.734 (0.657, 0.820)	
p-value	<0.00001 ^b	
IDFS at 84 months (%; 95% CI)	77.4 (75.6, 79.2)	70.9 (68.9, 72.8)
Distant Relapse Free Survival (DRFS)^a		
Number of patients with an event (n, %)	476 (17.0)	621 (22.0)
Hazard ratio (95% CI)	0.746 (0.662, 0.840)	
p-value	<0.00001 ^c	
DRFS at 84 months (%; 95% CI)	80.0 (78.2, 81.7)	74.9 (72.9, 76.7)
Overall Survival (OS)		
Number of patients with an event (n, %)	301 (10.7)	360 (12.7)
Hazard ratio (95% CI)	0.842 (0.722, 0.981) ^d	
p-value	0.02726 ^d	
OS at 84 months (%; 95% CI)	86.8 (85.1, 88.3)	85.0 (83.3, 86.4)

Abbreviation: CI = confidence interval.

Data cut-off date 15 July 2025

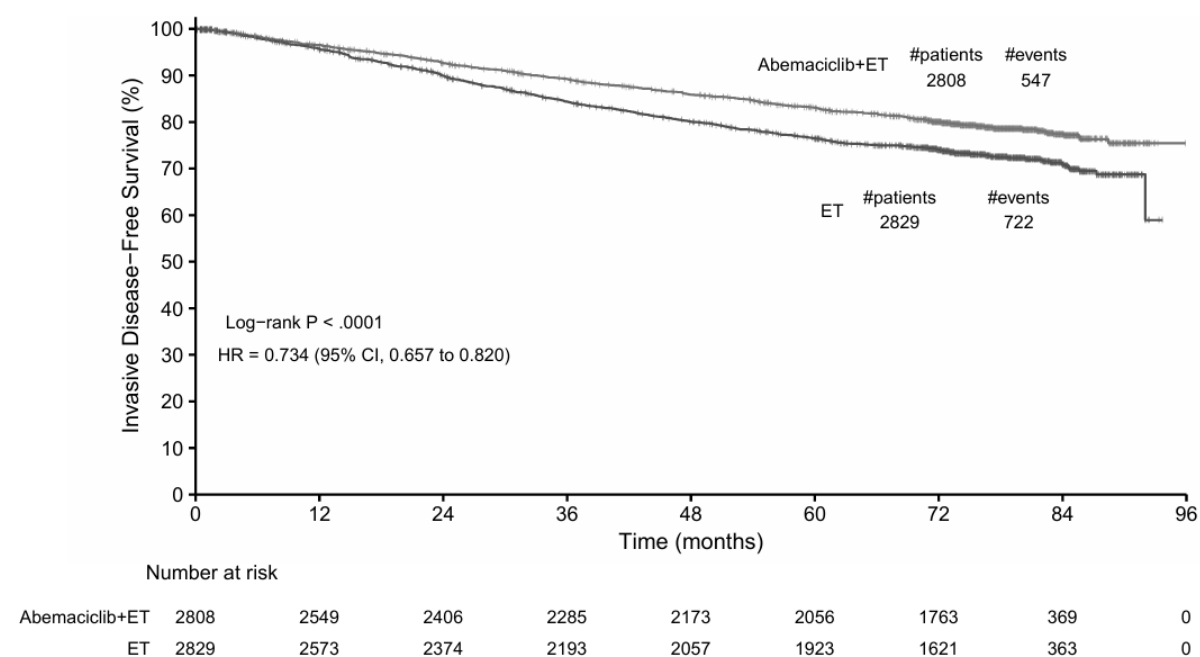
^a Distant relapse free survival is defined as the time from randomisation to distant recurrence or death from any cause, whichever occurs first.

^b Nominal p-values as statistical significance were achieved at Interim Analysis 2

^c Nominal p-values as DRFS was not alpha controlled

^d Stratified by IWRS Geographical Region, IWRS Prior Treatment, and IWRS Menopausal Status

Figure 1. monarchE: Kaplan-Meier plot of IDFS (Investigator assessment, intent-to-treat population)–at Final OS Analysis: VERZENIO plus Endocrine Therapy versus Endocrine Therapy Alone



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Advanced or metastatic breast cancer

Randomised phase 3 study MONARCH 3: VERZENIO in combination with aromatase inhibitors

The efficacy and safety of VERZENIO was evaluated in MONARCH 3, a randomised, double-blind, placebo-controlled phase 3 study in women with HR positive, HER2 negative locally advanced or metastatic breast cancer who had not received prior systemic therapy in this disease setting. Key exclusion criteria included visceral crisis or severe organ dysfunction, leptomeningeal carcinomatosis, inflammatory breast cancer, history of other cancer, clinical evidence or history of CNS metastases, pre-existing conditions resulting in clinically significant diarrhoea (such as Crohn's Disease, ulcerative colitis, or major resection of the small bowel and stomach), history of cardiovascular issues in the last 12 months, including history of venous thromboembolism.

Patients were randomised in a 2:1 ratio to receive VERZENIO 150 mg twice daily plus a non-steroidal aromatase inhibitor given daily at the recommended dose. The primary endpoint was investigator-assessed progression-free survival (PFS) evaluated according to RECIST 1.1; key secondary efficacy endpoints included objective response rate (ORR), clinical benefit rate (CBR) and overall survival (OS).

Patients were well matched for baseline demographics and prognostic characteristics between the abemaciclib and aromatase inhibitor arm (AI) and the placebo plus AI arm. The median age of patients enrolled was 63 years (range 32-88). Approximately 39% of patients had received chemotherapy and 44% had received antihormonal therapy in the (neo)adjuvant setting prior to their diagnosis of advanced breast cancer. The majority of patients (96%) had metastatic disease at baseline. Approximately 22% of patients had bone-only disease, and 53% patients had visceral metastases.

At the pre-planned interim analysis, the study met the primary endpoint demonstrating a statistically significant prolongation in PFS and a clinically meaningful treatment effect. Primary efficacy results are summarised in [Table 14](#) and [Figure 2](#).

Table 14. MONARCH 3: Summary of efficacy data (Investigator assessment, intent-to-treat population)

	VERZENIO plus aromatase inhibitor	Placebo plus aromatase inhibitor
Progression-free survival	N=328	N=165
Investigator assessment, number of events (%)	138 (42.1)	108 (65.5)
Median [months] (95% CI)	28.18 (23.51, NR)	14.76 (11.24, 19.20)
Hazard ratio (95% CI) and p-value	0.540 (0.418, 0.698), p=0.000002	
Independent radiographic review, number of events (%)	91 (27.7)	73 (44.2)
Median [months] (95% CI)	NR (NR, NR)	19.36 (16.37, 27.91)
Hazard ratio (95% CI) and p-value	0.465 (0.339, 0.636); p < 0.000001	
Objective response rate^b [%] (95% CI)	49.7 (44.3, 55.1)	37.0 (29.6, 44.3)
Duration of response [months] (95% CI)	27.39 (25.74, NR)	17.46 (11.21, 22.19)

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	VERZENIO plus aromatase inhibitor	Placebo plus aromatase inhibitor
Objective response for patients with measurable disease^a	N=267	N=132
Objective response rate ^b [%] (95% CI)	61.0 (55.2, 66.9)	45.5 (37.0, 53.9)
Complete response, (%)	3.4	0
Partial response, (%)	57.7	45.5
Clinical benefit rate^c (measurable disease) [%] (95% CI)	79.0 (74.1, 83.9)	69.7 (61.9, 77.5)

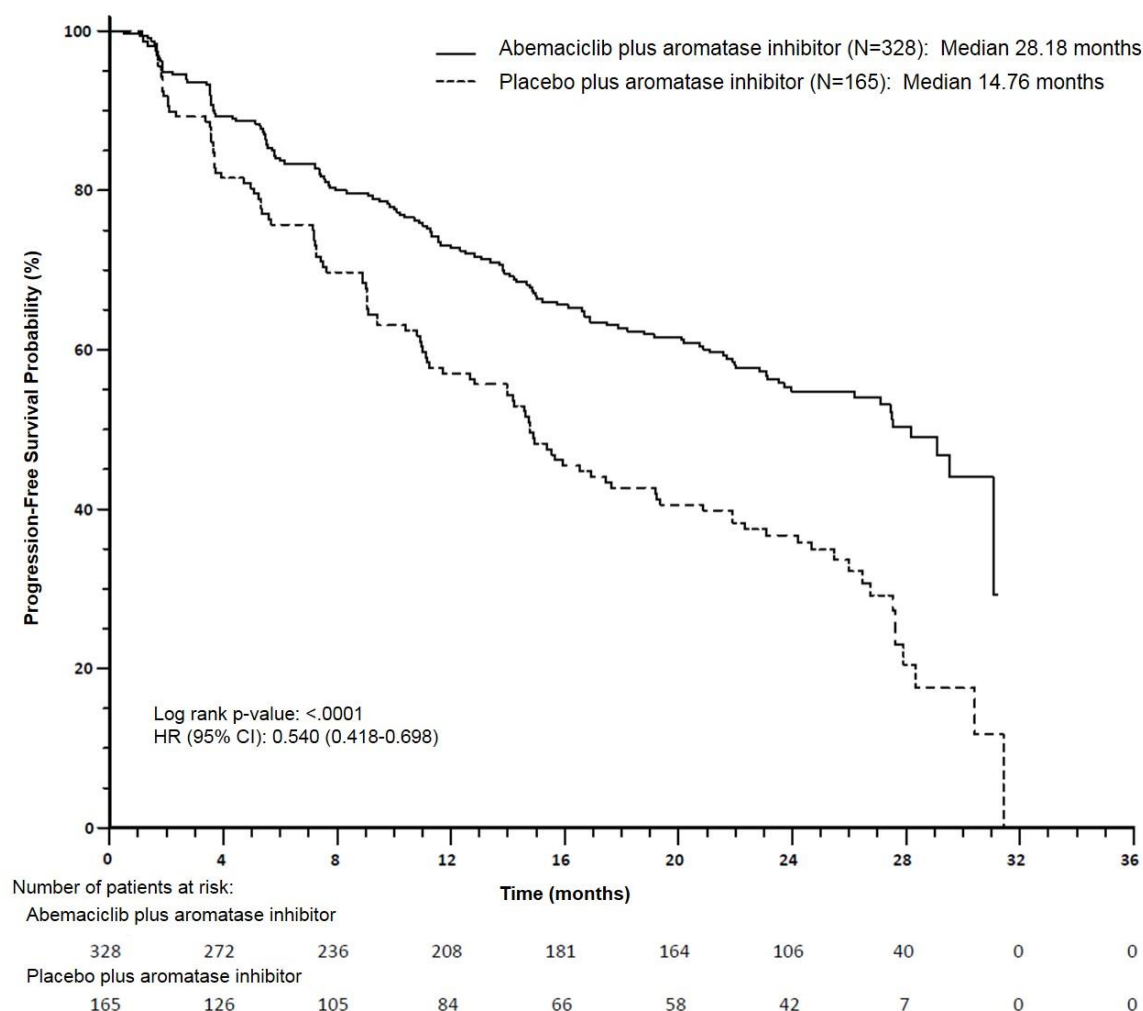
^a Measurable disease defined per RECIST version 1.1

^b Complete response + partial response

^c Complete response + partial response + stable disease for ≥ 6 months

N=number of patients; CI=confidence interval; NR=not reached.

Figure 2. MONARCH 3: Kaplan-Meier plot of progression-free survival (Investigator assessment, intent-to-treat population)



Progression-free survival (PFS) was significantly prolonged in the VERZENIO plus aromatase inhibitor (AI) arm, [Hazard Ratio [HR] of 0.540 [95% CI, 0.418 to 0.698]]; median PFS was 28.18 months in the VERZENIO plus AI arm and was 14.76 months in the placebo plus AI arm. These results correspond to a clinically meaningful reduction in the risk of disease progression or death of 46% for patients treated with abemaciclib plus an aromatase inhibitor.

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Overall survival was not mature at the final PFS analysis (93 events observed across the two arms). The HR was 1.057 (95% CI: 0.683, 1.633), $p=0.8017$.

A series of prespecified subgroup PFS analyses showed consistent results across patient subgroups including age (<65 or ≥ 65 years), disease site, disease setting (de novo metastatic vs recurrent metastatic vs locally advanced recurrent), presence of measurable disease, progesterone receptor status, and baseline ECOG performance status. A reduction in the risk of disease progression or death was observed in patients with visceral disease, (HR of 0.567 [95% CI: 0.407, 0.789]), median PFS 21.6 months versus 14.0 months; in patients with bone-only disease (HR 0.565, [95% CI: 0.306, 1.044]); and in patients with measurable disease (HR 0.517, [95% CI: 0.392, 0.681]).

Randomised phase 3 study MONARCH 2: VERZENIO in combination with fulvestrant

The efficacy and safety of VERZENIO was evaluated in MONARCH 2, a randomised, double-blind, placebo-controlled phase 3 study in women with HR positive, HER2 negative locally advanced or metastatic breast cancer. Key exclusion criteria included visceral crisis or severe organ dysfunction, leptomeningeal carcinomatosis, inflammatory breast cancer, history of other cancer, clinical evidence or history of CNS metastases, pre-existing conditions resulting in clinically significant diarrhoea (such as Crohn's Disease, ulcerative colitis, or major resection of the small bowel and stomach), history of cardiovascular issues in the last 12 months, including history of venous thromboembolism.

Patients were randomised in a 2:1 ratio to receive VERZENIO 150 mg twice daily plus fulvestrant 500 mg at intervals of one month, with an additional 500 mg dose given two weeks after the initial dose, versus placebo plus fulvestrant alone according to the same schedule. The primary endpoint was investigator-assessed PFS evaluated according to RECIST 1.1; key secondary efficacy endpoints included objective response rate (ORR), clinical benefit rate (CBR) and overall survival (OS).

Patients were well matched for baseline demographics and prognostic characteristics between the abemaciclib plus fulvestrant arm and the placebo plus fulvestrant arm. The median age of patients enrolled was 60 years (range, 32-91 years). In each treatment arm the majority of patients were white and had not received chemotherapy or any prior endocrine therapy for metastatic disease. 17% of patients were pre/perimenopausal. Approximately 56% patients had visceral metastases. Approximately 25% patients had primary resistance to endocrine therapy as per ESMO International Consensus Guidelines for Advanced Breast Cancer, with the majority of patients having secondary resistance.

The study met the primary endpoint demonstrating a statistically significant prolongation in PFS and a clinically meaningful treatment effect. Primary efficacy results are summarised in [Table 15](#) and [Figure 3](#).

Table 15. MONARCH 2: Summary of efficacy data (Investigator assessment, intent-to-treat population)

	VERZENIO plus fulvestrant	Placebo plus fulvestrant
Progression-free survival	N=446	N=223
Investigator assessment, number of events (%)	222 (49.8)	157 (70.4)

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	VERZENIO plus fulvestrant	Placebo plus fulvestrant
Median [months] (95% CI)	16.4 (14.4, 19.3)	9.3 (7.4, 12.7)
Difference in PFS (months)	7.2	
Hazard ratio (95% CI) and p-value	0.553 (0.449, 0.681), p=0.0000001	
Independent radiographic review, number of events (%)	164 (36.8)	124 (55.6)
Median [months] (95% CI)	22.4 (18.3, NR)	10.2 (5.8, 14.0)
Hazard ratio (95% CI) and p-value	0.460 (0.363, 0.584); p <.0000001	
Objective response rate^a [%] (95% CI)	35.2 (30.8, 39.6)	16.1 (11.3, 21.0)
Duration of response [months] (95%CI)	NR (18.05, NR)	25.6 (11.9, 25.6)
Objective response for patients with measurable disease	N=318	N=164
Objective response rate ^a [%] (95% CI)	48.1 (42.6, 53.6)	21.3 (15.1, 27.6)
Complete response, (%)	3.5	0
Partial response, (%)	44.7	21.3
Disease control rate^b (measurable disease) [%] (95% CI)	82.4 (78.2, 86.6)	72.6 (65.7, 79.4)
Clinical benefit rate^c (measurable disease)	73.3 (68.4, 78.1)	51.8 (44.2, 59.5)

^a Complete response + partial response

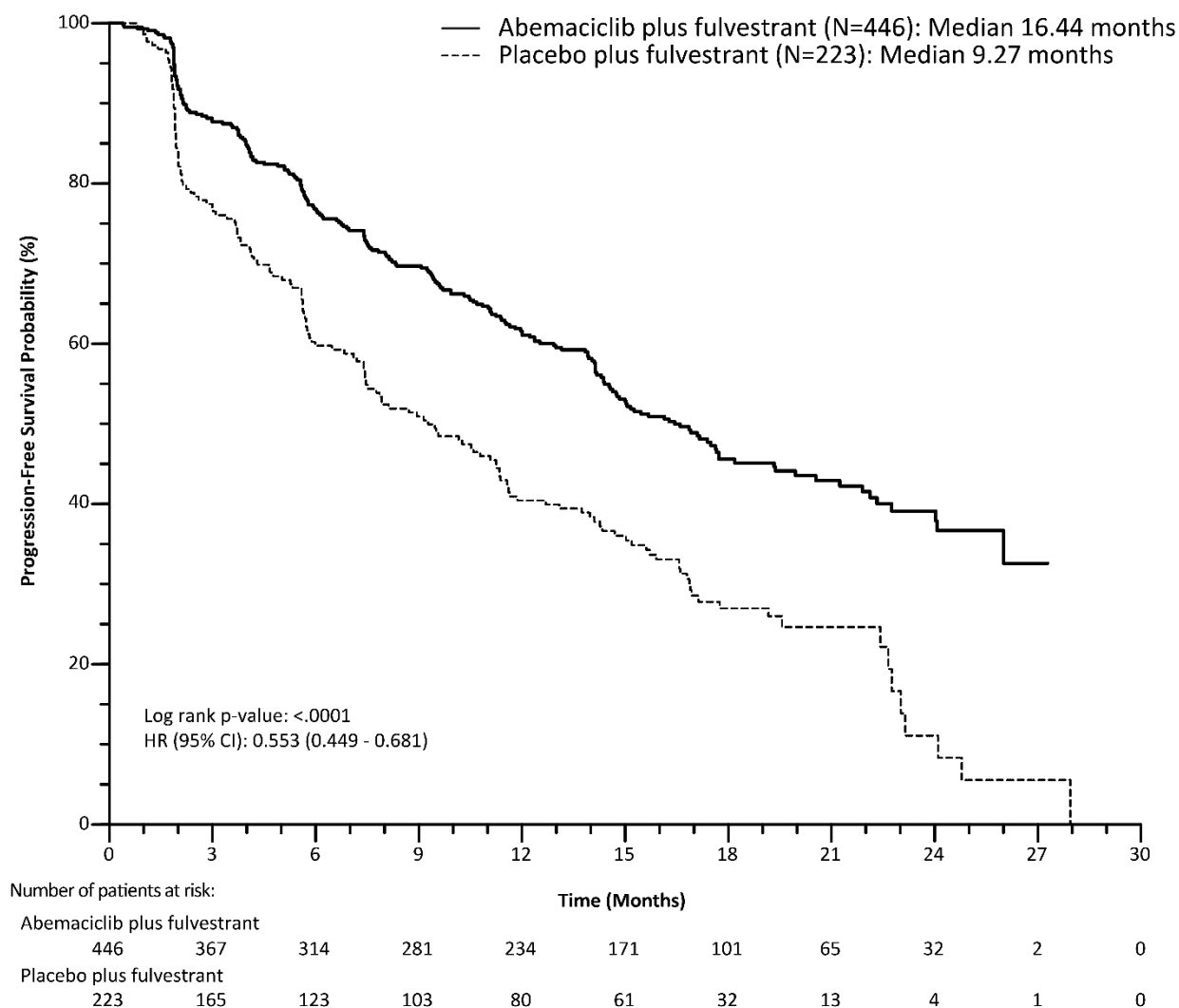
^b Complete response + partial response + stable disease

^c Complete response + partial response + stable disease for ≥ 6 months

N=number of patients; CI=confidence interval

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Figure 3. MONARCH 2: Kaplan-Meier plot of progression-free survival (Investigator assessment, intent-to-treat population)



Median PFS was significantly prolonged in the VERZENIO plus fulvestrant arm (HR of 0.553 [95% CI 0.449, 0.681]); median PFS was 16.4 months versus 9.3 months in the placebo plus fulvestrant arm. These results correspond to a clinically meaningful reduction in the risk of disease progression or death of 44.7% and a 7.2 month improvement in median PFS for patients treated with VERZENIO plus fulvestrant. Early and sustained separation by treatment arm was apparent beginning at 8 weeks. VERZENIO plus fulvestrant prolonged progression-free survival with neither a clinically meaningful or significant detriment to health-related quality of life.

The addition of VERZENIO to fulvestrant significantly delayed the time to post-discontinuation chemotherapy, hazard ratio 0.651 (95% CI: 0.502, 0.845). The median time to chemotherapy for the abemaciclib arm was not yet reached and for the placebo plus fulvestrant arm was 26 months.

A series of prespecified subgroup PFS analyses were performed based on prognostic factors and baseline characteristics to confirm consistency of the treatment effect. A reduction in the risk of disease progression or death in favour of the VERZENIO plus fulvestrant arm was observed in all patient subgroups. Consistent results were observed across patient subgroups including age (<65 or ≥65 years), race, geographic region, disease site, endocrine therapy resistance, presence of measurable disease, progesterone receptor status, and menopausal status. A reduction in the risk

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of disease progression or death was evident in patients with visceral disease, (HR of 0.481 [95% CI: 0.369, 0.627], median PFS 14.7 months versus 6.5 months); in patients with bone-only disease (HR of 0.543 [95% CI: 0.355, 0.833]); patients with measurable disease (HR of 0.523 [95% CI: 0.412, 0.644]). In patients who were pre/perimenopausal, the hazard ratio was 0.415 (95% CI: 0.246, 0.698); in patients who were progesterone receptor negative, the HR was 0.509 [95% CI: 0.325, 0.797]).

In the population of 44 patients who presented de novo with locally advanced or metastatic disease, and had not received any prior endocrine therapy, the addition of VERZENIO to fulvestrant reduced the risk of disease progression or death in this population by 54.6% (HR of 0.454 [95% CI: 0.179, 1.154]).

Overall survival (OS) analysis in the ITT population showed a statistically significant improvement in patients receiving VERZENIO plus fulvestrant compared with those receiving placebo plus fulvestrant. The overall survival results are summarised in [Table 16](#) and [Figure 4](#).

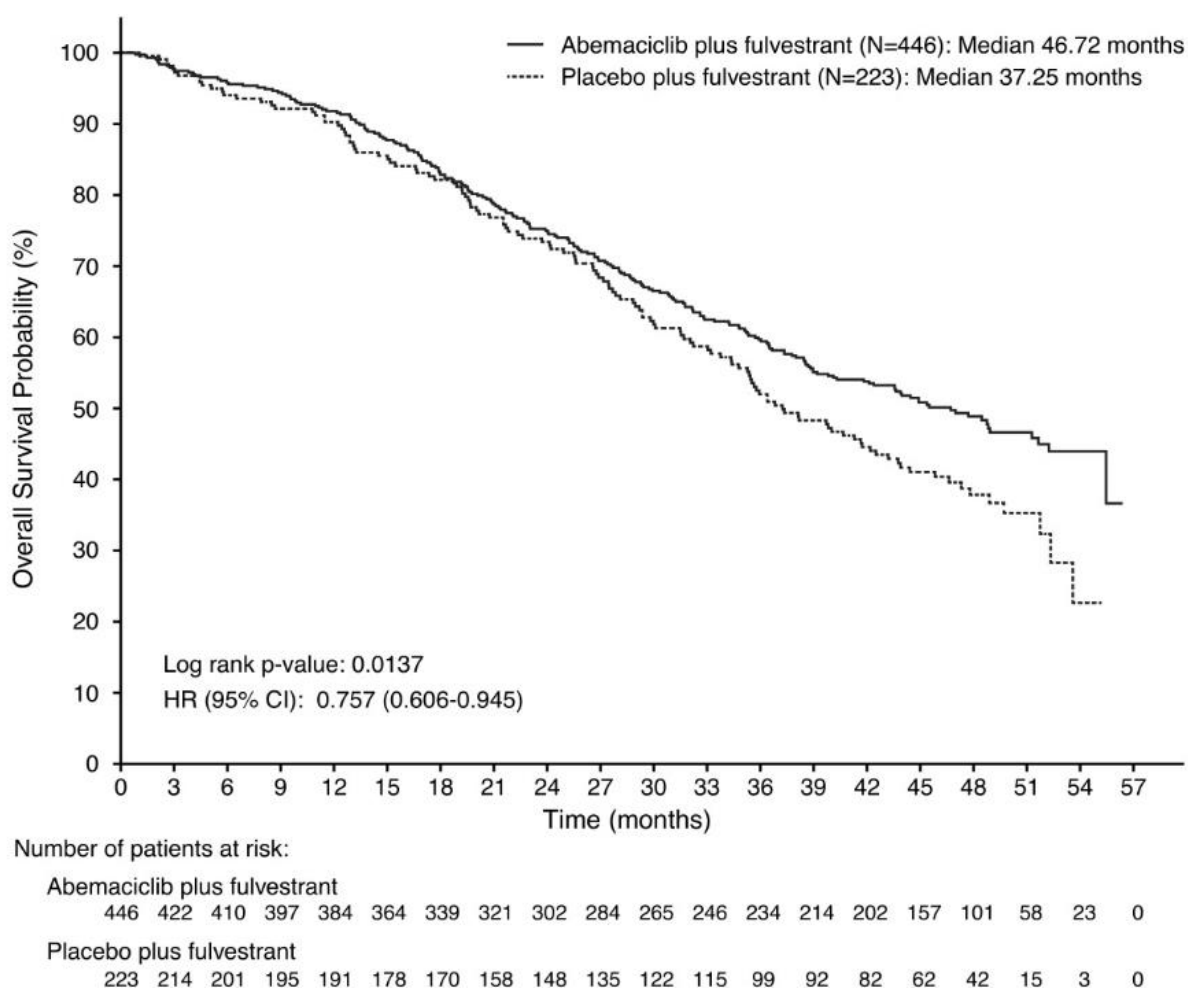
Table 16. MONARCH 2: Summary of overall survival data (Intent-to-treat population)

	VERZENIO plus fulvestrant	Placebo plus fulvestrant
Overall survival	N = 446	N = 223
Number of events (n, %)	211 (47.3)	127 (57.0)
Median OS [months] (95 % CI)	46.7 (39.2, 52.2)	37.3 (34.4, 43.2)
Hazard ratio (95 % CI)	0.757 (0.606, 0.945)	
p-value	0.0137	

N = number of patients; CI = confidence interval; OS = overall survival

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Figure 4. MONARCH 2: Kaplan-Meier plot of overall survival (Intent-to-treat population)



Analyses for OS by stratification factors showed OS HR of 0.675 (95 % CI: 0.511, 0.891) in patients with visceral disease, and 0.686 (95 % CI: 0.451, 1.043) in patients with primary endocrine resistance.

5.2 PHARMACOKINETIC PROPERTIES

Absorption

Abemaciclib absorption is slow, with a median T_{max} of 8.0 hours. The absolute bioavailability of abemaciclib is 45% (90% confidence interval: 40-51%). In the therapeutic dose range of 50-200 mg, the increase in plasma exposure (AUC) and C_{max} is dose proportional. Steady state was achieved within 5 days following repeated twice daily dosing, and abemaciclib accumulated with a geometric mean accumulation ratio of 3.7 (58% CV) and 5.8 (65% CV) based on C_{max} and AUC, respectively.

Distribution

Abemaciclib was highly bound to plasma proteins in humans (mean bound fraction was approximately 96-98%), and the binding was independent of concentration from 152 ng/mL to 5066 ng/mL. Abemaciclib binds to both human serum albumin and alpha-1-acid glycoprotein. The geometric mean systemic volume of distribution is approximately 747 L (68.6% CV). In

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patients with advanced cancer, concentrations of abemaciclib and its active metabolites M2 and M20 in cerebrospinal fluid are comparable to unbound plasma concentrations.

Biotransformation

Hepatic metabolism is the main route of clearance for abemaciclib. Abemaciclib is metabolized to several metabolites primarily by cytochrome P450 (CYP) 3A, with formation of N-desethyl abemaciclib (M2) representing the major metabolism pathway. Additional metabolites include hydroxyabemaciclib (M20), hydroxy-N-desethylabemaciclib (M18), and an oxidative metabolite (M1). Metabolites N-desethylabemaciclib (M2) and hydroxyabemaciclib (M20) are active with similar potency as abemaciclib.

Elimination

The geometric mean hepatic clearance (CL) of abemaciclib was 21.8 L/h (39.8% CV), and the mean plasma elimination half-life for abemaciclib in patients was 24.8 hours (52.1% CV). After a single oral dose of [¹⁴C]-abemaciclib, approximately 81% of the dose was excreted in faeces and 3.4% excreted in urine. The majority of the dose eliminated in faeces was metabolites.

Use in hepatic impairment

Abemaciclib is metabolised in the liver. Mild and moderate hepatic impairment (Child-Pugh classification), did not have a clinically relevant effect on abemaciclib exposure. In subjects with severe hepatic impairment (Child-Pugh classification), total abemaciclib unbound exposure increased 2.69-fold, and the abemaciclib half-life increased from 24 to 55 hours. Reduce the abemaciclib dosing frequency to once daily in patients with severe hepatic impairment.

Use in renal impairment

Abemaciclib and its metabolites are not significantly cleared renally. Dose adjustment is not necessary in patients with mild or moderate renal impairment. There are no data in patients with severe renal impairment, end stage renal disease, or in patients on dialysis.

Use in the elderly

Age had no effect on the exposure of abemaciclib in a population pharmacokinetic analysis in patients with cancer (135 males and 859 females; age range 24-91 years; and body weight range 36-175 kg).

5.3 PRECLINICAL SAFETY DATA

Genotoxicity

Abemaciclib was not mutagenic in a bacterial reverse mutation (Ames) assay and did not induce structural chromosomal aberrations in the in vitro chromosome aberration assay with human lymphocytes, or the in vivo rat micronucleus test. Metabolites M2 and M20 were not mutagenic in the Ames assay, did not induce structural chromosomal aberrations in Chinese Hamster Ovary cells in the in vitro chromosome aberration assay, and were not clastogenic in the rat bone marrow micronucleus assay.

Carcinogenicity

In male rats treated for 95 weeks with abemaciclib at $\geq 1\text{mg/kg/day}$ (less than or similar to clinical exposure based on AUC), abemaciclib caused interstitial (Leydig) cell hyperplasia and

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benign adenomas in testes. It is unknown if this effect will translate to humans. There were no neoplastic findings in mice or female rats.

6 PHARMACEUTICAL PARTICULARS

6.1 LIST OF EXCIPIENTS

<u>Tablet cores</u>	<u>Film coating</u>
croscarmellose sodium	polyvinyl alcohol (E1203)
lactose monohydrate	titanium dioxide (E171)
microcrystalline cellulose	macrogol 3350 (E1521)
silicon dioxide	purified talc (E553b)
sodium stearyl fumarate	iron oxide yellow (E172) [50 mg and 150 mg tablets only]
	iron oxide red (E172) [50 mg tablet only]

6.2 INCOMPATIBILITIES

In absence of compatibility studies, this medicine must not be mixed with other medicines.

6.3 SHELF LIFE

3 years.

6.4 SPECIAL PRECAUTIONS FOR STORAGE

Store below 30°C.

6.5 NATURE AND CONTENTS OF CONTAINER

VERZENIO is supplied in PVC/PE/PCTFE blister packs sealed with aluminum foil lidding.

Pack sizes:

50 mg, 100 mg, 150 mg: 56 tablets

6.6 SPECIAL PRECAUTIONS FOR DISPOSAL

Any unused medicine or waste material should be disposed of in accordance with local requirements.

7 MEDICINE SCHEDULE

Prescription

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8 SPONSOR

Eli Lilly and Company (NZ) Limited
PO Box 109 197
Newmarket
Auckland 1149
Phone: 0800 500 056
Web: www.lilly.co.nz

9 DATE OF FIRST APPROVAL

30 November 2023

10 DATE OF REVISION OF THE TEXT

28 July 2026

SUMMARY TABLE OF CHANGES

Section Changed	Summary of new information
All	Formatting
5.1	Update monarchE with final Overall Survival (OS) data (7years)